

SWAG: Scale-Wise Adaptive Gating for Rotary Position Embeddings

James Baker*

July 31, 2026

Abstract

Rotary position embeddings (RoPE) give every token in every context the same fixed bank of frequency channels: a delimiter resolving local syntax and a pronoun searching for a distant antecedent are handed identical positional geometry. Recent work makes the geometry itself input-dependent, which recovers some flexibility but gives up strict relativity, the property that an attention logit depends on positions only through their difference. We propose SWAG (scale-wise adaptive gating), which keeps the geometry fixed and gates its *use*. Each query token emits, per head, a sigmoid gate over RoPE’s frequency channels that scales only the positional correction each channel contributes to the attention logit. Because the gates depend on the query token alone and the geometry inside the correction depends on the relative offset alone, shift consistency is preserved exactly; setting all gates to one recovers standard RoPE and setting them to zero recovers a content-only comparison. We give an exact reformulation that computes the gated logit as a single inner product in a doubled head dimension, so the method runs on fused attention kernels with no per-channel score materialization. We compare six matched positional variants (no positional encoding, RoPE, learned per-head frequencies, a re-implementation of context-aware rotary frequencies, SWAG, and a channel-gating ablation) at approximately 5M parameters under a fixed compute budget on byte-level enwik8, on synthetic offset-recall and copy probes, and at 1x/2x/4x the training context. On enwik8, SWAG attains the best bits per character of the six at the training context (1.479 against 1.499 for RoPE, consistent across seeds) and degrades less than RoPE at every longer context (3.25 against 3.58 at 4x), though learned per-head frequencies extrapolate best on natural text (2.31). The synthetic probes show the price of that extrapolation: the variants that modify the frequency bank lose exact offset recall at longer sequences (down to 0.50–0.73 accuracy), while SWAG matches standard RoPE’s perfect generalization everywhere. SWAG is the only variant tested that improves on RoPE without giving up any of RoPE’s relative-position robustness. Inspecting the learned gates shows systematic structure: digit tokens suppress the most local channels, middle layers converge on a shared mid-band scale modulated by character class, and mean gate values decline with depth while the learned per-head positional scale rises above its initialization. Code and all run logs are public.

1 Introduction

Rotary position embeddings (Su et al., 2021) have become the default positional mechanism in decoder-only language models. RoPE rotates each two-dimensional pair of a head’s query and key by an angle proportional to absolute position, with a geometrically spaced frequency bank $\theta_i = b^{-2i/d_h}$. The resulting attention logit depends on positions only through the offset $n - m$, and the $I = d_h/2$ channels form a multi-scale basis over relative distance: the fastest channels distinguish adjacent positions, the slowest vary on the scale of the whole context.

*Independent researcher. Correspondence: james.baker1628@gmail.com.

The bank is fixed and shared. Every token, in every context, at every layer, is compared using the same weighting of these scales. That is a strong assumption. A token opening a bracket, a token continuing a word, and a token that must retrieve a referent hundreds of positions back plausibly need different parts of the distance spectrum; a single fixed weighting must compromise between them, and the compromise is baked into the architecture rather than learned per token. The same rigidity shows up at test time as brittleness beyond the training context, which is why a body of work retrofits the frequency bank after training (Peng et al., 2024) or replaces it with a monotone distance penalty (Press et al., 2022).

A recent line of work makes the positional geometry itself a function of the input. CARoPE (Veisi et al., 2025) generates head-specific frequency patterns from token embeddings; Selective RoPE (Movahedi et al., 2025) makes rotation angles input-dependent inside gated linear attention; CoPE (Golovneva et al., 2024) lets the model decide which tokens increment the position counter at all. These methods buy expressiveness by giving up strict relativity. Once the query at position m uses frequency θ_m and the key at n uses θ_n , the phase difference $n\theta_n - m\theta_m$ is no longer a function of $n - m$, so translating a passage within the context changes its internal attention pattern. Relativity is not a formality: it is the reason RoPE transfers across sequence lengths and positions at all.

We take the opposite decomposition. Keep RoPE’s geometry exactly as it is, and let each token modulate *how much it uses* each scale. Note that a channel’s RoPE contribution splits into a content term $q_i^\top k_i$ and a positional correction $q_i^\top R((n - m)\theta_i)k_i - q_i^\top k_i$. SWAG gates only the correction: the query token at position m emits, per head, a sigmoid gate $g_{m,h,i} \in (0, 1)$ over the I channels, and channel i ’s correction enters the logit scaled by $\lambda_h g_{m,h,i}$, where λ_h is a learned per-head scalar. Because the gate is a function of the query token only and the correction is a function of $n - m$ only, shift consistency is preserved exactly. The endpoints are interpretable: gates at one give standard RoPE, gates at zero give a content-only comparison in that channel, and intermediate values interpolate between them without ever changing what “distance $n - m$ ” means.

The naive form of this logit appears to require materializing $O(T^2 I)$ per-channel scores. It does not: collecting terms yields an exact expression as a single inner product between an effective query and key of doubled head dimension, compatible with fused attention kernels, with values and attention outputs unchanged in size.

Contributions.

- **Method.** SWAG, a per-token per-head gating of RoPE’s positional corrections that leaves the relative, shift-consistent geometry exactly intact and contains RoPE and no-positional-encoding as the two gate endpoints (Section 3.2).
- **Exact fast form.** A reformulation computing the gated logit as one inner product in $2d_h$ dimensions, verified against the $O(T^2 I)$ reference to 10^{-10} in float64, so the method costs a gate projection and a wider $QK^\top$ contraction rather than a new attention kernel (Section 3.3).
- **Controlled comparison.** A compute- and parameter-matched study of six positional variants (NOPE, ROPE, LEARNED-FREQ, CAROPE, SWAG, GATED-SIMPLE) on byte-level enwik8, on offset-recall and copy probes, and at 1x/2x/4x training context, at a scale where every variant sees an identical budget (Section 4).
- **Gate interpretability.** An analysis of what the learned gates encode, aggregating mean gate value by character class, layer, head, and frequency index, plus the learned per-head λ_h (Section 5).

Findings are reported in Section 5; all numbers there are produced by the runs described in Section 4 and none are estimated or extrapolated.

2 Related work

Fixed relative position encodings. RoPE (Su et al., 2021) encodes absolute position by rotating two-dimensional query and key pairs, so that the attention logit depends only on the relative offset. Earlier relative schemes add a learned scalar bias per bucketed distance, as in the T5 relative attention bias (Raffel et al., 2020); ALiBi (Press et al., 2022) replaces the learned table with a fixed per-head linear penalty in distance, trading expressiveness for length extrapolation. Kazemnejad et al. (2023) find that decoder-only transformers with no explicit positional encoding at all (NOPE) can outperform explicit schemes on length generalization, since causal masking already leaks position; we include NOPE as a baseline for that reason. YaRN (Peng et al., 2024) extends a trained RoPE model’s context by rescaling the frequency bank non-uniformly, with an interpolation strength that varies by wavelength. What all of these share with RoPE is that the mapping from relative distance to logit contribution is fixed at training time and identical for every token.

Input-dependent positional geometry. A recent line of work makes the geometry itself depend on the input. CARoPE (Veisi et al., 2025) generates token- and head-specific frequency patterns from token embeddings, so the rotation angle applied at a position is a function of the token sitting there. Selective RoPE (Movahedi et al., 2025) makes rotation angles input-dependent within gated linear and softmax attention, allowing rotation by arbitrary learned angles. CoPE (Golovneva et al., 2024) goes further and makes the position index itself content-dependent, incrementing the counter only on tokens the model selects, which lets attention address the j -th sentence rather than the j -th token. These methods gain flexibility at the cost of strict relativity: when the query at m and the key at n carry different frequencies or different position counters, the logit is no longer a function of $n - m$ alone, and the same span placed at a different offset in the context is compared differently. We include a re-implementation of the CARoPE mechanism (Section 4) as the representative of this family, and note explicitly that it is our approximation of the published method rather than the authors’ code.

Gated positional information. Closest in spirit to our approach is GAPE (Ali et al., 2026), which introduces query-dependent gating on top of a retained RoPE backbone, modulating how strongly distance suppresses attention so that irrelevant context is contracted while salient distant tokens survive. The distinction is what gets gated. GAPE gates an aggregate distance-decay term, a scalar-valued control over “how far should I look”; SWAG gates the I frequency channels individually, a vector-valued control over “which scales of relative structure should I use”, of which uniform distance sensitivity is one special case. Gating individual bands is also what makes the endpoints clean: channel-wise gates at one recover RoPE exactly and at zero recover a content-only comparison in that band, so the method interpolates within a family that already contains both baselines.

Positioning. The three families differ in what they allow the input to control. CARoPE and Selective RoPE let the input change the positional geometry, sacrificing the relative property. GAPE keeps the geometry and gates an aggregate decay. SWAG keeps the geometry and gates the *use* of each individual relative-frequency band by the querying token, which preserves shift consistency exactly (Section 3.2) while still letting different tokens attend on different scales. It is also strictly

more expressive than the LEARNED-FREQ baseline we compare against, in which per-head frequencies are learned but remain constant across tokens.

3 Method

3.1 Preliminaries: rotary position embeddings

Let $x_m \in \mathbb{R}^d$ denote the residual-stream representation of the token at position m . For attention head h with head dimension d_h , queries and keys are $q_{m,h} = W_Q^{(h)} x_m$ and $k_{n,h} = W_K^{(h)} x_n$. RoPE (Su et al., 2021) partitions each head’s query and key into $I = d_h/2$ two-dimensional pairs $q_{m,h,i} \in \mathbb{R}^2$, $i \in \{0, \dots, I-1\}$, and rotates pair i at position m by the angle $m\theta_i$, where the fixed frequency bank

$$\theta_i = b^{-2i/d_h}, \quad b = 10,000 \quad (1)$$

spans wavelengths from 2π positions ($i=0$, the most local channel) up to $2\pi b$ positions ($i=I-1$, the longest-range channel). Writing $R(\alpha) \in SO(2)$ for the rotation by angle α , the attention logit becomes

$$s_{m,n,h}^{\text{RoPE}} = \frac{1}{\sqrt{d_h}} \sum_{i=0}^{I-1} q_{m,h,i}^\top R((n-m)\theta_i) k_{n,h,i}, \quad (2)$$

which depends on positions only through the relative displacement $n-m$: the identity $R(m\theta_i)^\top R(n\theta_i) = R((n-m)\theta_i)$ makes RoPE relative, shift-consistent, and smooth in $n-m$, with the I channels forming a fixed multi-scale basis over relative distance.

Every token and every context, however, uses this basis with the same fixed weighting. A delimiter token resolving local syntax and a pronoun searching a long context for its antecedent receive identical positional geometry.

3.2 SWAG: gating the positional correction per frequency channel

Our goal is to let each query token choose *how much each positional scale matters to it* without letting it redefine the notion of position itself. The key observation is that each channel’s RoPE logit contribution decomposes into a content term and a purely positional correction:

$$\underbrace{q_{m,h,i}^\top R((n-m)\theta_i) k_{n,h,i}}_{\text{RoPE term}} = \underbrace{q_{m,h,i}^\top k_{n,h,i}}_{\text{content } c_{m,n,h,i}} + \underbrace{\left[q_{m,h,i}^\top R((n-m)\theta_i) k_{n,h,i} - q_{m,h,i}^\top k_{n,h,i} \right]}_{\text{positional correction } p_{m,n,h,i}}. \quad (3)$$

SWAG gates only the correction $p_{m,n,h,i}$. Each query token emits, per head, a gate vector over the I frequency channels,

$$g_{m,h} = \sigma(W_g^{(h)} \text{RMSNorm}(x_m) + b_g^{(h)}) \in (0, 1)^I, \quad (4)$$

and the logit becomes

$$s_{m,n,h} = \frac{1}{\sqrt{d_h}} \left[q_{m,h}^\top k_{n,h} + \lambda_h \sum_{i=0}^{I-1} g_{m,h,i} p_{m,n,h,i} \right], \quad (5)$$

where λ_h is a learned per-head scalar separating *which* scales matter (the gates) from *how much position matters overall* in head h . At $g_{m,h,i} = 1$, $\lambda_h = 1$ Eq. 5 is exactly standard RoPE (Eq. 2); at $g_{m,h,i} = 0$ channel i contributes a purely content-based comparison, position ignored. Because the gates depend only on the query token and the positional geometry inside p depends only on $n-m$, shift consistency is preserved exactly: translating all positions by a constant leaves every logit unchanged. This is the property that token-conditioned *frequencies* (Veisi et al., 2025) give up, since a phase difference $n\theta_n - m\theta_m$ with token-dependent θ is not a function of $n-m$ alone.

Initialization. We initialize $b_g = 2$, so $g \approx 0.88$ and the model starts close to standard RoPE, learning to *turn channels down* selectively; $\lambda_h = 1$; and $W_g \sim \mathcal{N}(0, 0.02^2)$. The gate input uses a parameter-free RMS normalization of the residual stream.

3.3 Exact computation without materializing per-channel scores

Naively, Eq. 5 requires $O(T^2I)$ per-channel score materialization. It can instead be computed exactly as a single inner product in a doubled head dimension. Substituting Eq. 3 into Eq. 5 and collecting terms, with $c_{m,h,i} \equiv \lambda_h g_{m,h,i}$:

$$s_{m,n,h} = \frac{1}{\sqrt{d_h}} \sum_{i=0}^{I-1} \left[(1 - c_{m,h,i}) q_{m,h,i}^\top k_{n,h,i} + c_{m,h,i} (R(m\theta_i) q_{m,h,i})^\top (R(n\theta_i) k_{n,h,i}) \right]. \quad (6)$$

Defining the effective queries and keys

$$\tilde{q}_{m,h} = [(1 - c_{m,h}) \odot q_{m,h} \parallel c_{m,h} \odot R(m\theta) q_{m,h}] \in \mathbb{R}^{2d_h}, \quad (7)$$

$$\tilde{k}_{n,h} = [k_{n,h} \parallel R(n\theta) k_{n,h}] \in \mathbb{R}^{2d_h}, \quad (8)$$

(with $\odot$ broadcasting each per-pair gate over its two dimensions) gives $s_{m,n,h} = \tilde{q}_{m,h}^\top \tilde{k}_{n,h} / \sqrt{d_h}$, directly compatible with fused attention kernels; the attention scale remains $1/\sqrt{d_h}$, as the doubling is an implementation device rather than an increase in effective dimensionality. The overhead relative to standard RoPE is the gate projection ($d \times HI$ parameters per layer) and a doubling of the $QK^\top$ contraction width; values, and hence the attention output, are unchanged in size. Our tests verify this fast form against the $O(T^2I)$ reference logit to 10^{-10} in float64.

3.4 An ablation: gating channels rather than corrections

A cheaper alternative scales each *rotated query pair* directly, $\tilde{q}_{m,h,i} = a_{m,h,i} R(m\theta_i) q_{m,h,i}$ with $a_{m,h,i} \in (0, 2)$, requiring no dimension doubling. This entangles content and position: setting $a_{m,h,i} = 0$ removes channel i 's content comparison along with its positional correction, so "ignore position here" and "ignore this subspace here" cannot be expressed separately. We evaluate this variant (GATED-SIMPLE) as an ablation to isolate the value of gating the correction specifically.

4 Experimental setup

The comparison is between positional mechanisms, so everything else is held fixed: identical architecture, identical optimizer, identical data order given a seed, identical step budget. Only the positional module inside attention changes.

4.1 Variants

Six variants share the interface $(q, k, x, \text{positions}) \mapsto (\tilde{q}, \tilde{k}, \text{scale})$ and are swapped into the same attention block.

nope No positional encoding: $\tilde{q} = q$, $\tilde{k} = k$. Causal masking is the only source of position (Kazemnejad et al., 2023).

rope Standard RoPE (Su et al., 2021) with the fixed bank $\theta_i = 10000^{-2i/d_h}$, shared across heads and layers.

Table 1: Parameter counts at the main tier ($d = 256$, 6 layers, 8 heads, $d_h = 32$, byte vocabulary 256, tied embeddings). Spread from smallest to largest is 4.1%.

Variant	Parameters	vs. RoPE
NOPE	4,801,280	+0.00%
ROPE	4,801,280	+0.00%
LEARNED-FREQ	4,802,048	+0.02%
CAROPE	4,998,656	+4.11%
GATED-SIMPLE	4,998,656	+4.11%
SWAG	4,998,704	+4.11%

learned-freq Per-head learned frequencies: a parameter $\log \theta \in \mathbb{R}^{H \times I}$ initialized at the RoPE bank and trained. Frequencies vary by head but not by token, so relativity is retained.

carope Our re-implementation of token-conditioned frequencies (Veisi et al., 2025): per token and head, $\theta_{m,h,i} = 2\theta_i \cdot \sigma(W_f \text{RMSNorm}(x_m) + b_f)_{h,i}$ with $b_f = 0$ so the multiplier starts at 1 and the variant starts at standard RoPE. The query at m rotates by $m\theta_m$ and the key at n by $n\theta_n$, which breaks shift consistency by construction. This is an approximation of the published method, not the authors’ implementation, and is included to represent the input-dependent-geometry family under matched conditions.

swag The method of Section 3: per-token per-head sigmoid gates over the I positional corrections, gate bias initialized to +2 ($g \approx 0.88$), $\lambda_h = 1$, gate projection $\mathcal{N}(0, 0.02^2)$, computed by the exact doubled-dimension form.

gated-simple Ablation of Section 3.4: the whole rotated query pair is scaled by $a_{m,h,i} = 2\sigma(\cdot) \in (0, 2)$, bias initialized to 0 so $a \approx 1$. Content and position are gated together, and no dimension doubling is needed.

Each layer carries its own gate (or frequency) projection; gates are cheap relative to the MLP. Gate and frequency projections are initialized $\mathcal{N}(0, 0.02^2)$ throughout.

4.2 Parameter matching

Table 1 gives exact parameter counts at the main (enwik8) tier. The spread between the smallest and largest variant is 4.1%, driven by the per-layer gate projection ($d \times HI$ weights, 197,376 parameters over six layers); the gated variants are the larger ones, so any advantage they show is not obtained at a parameter deficit but does carry a small parameter surplus, which we note when interpreting results. Step budget and batch size are identical across variants, so wall-clock and FLOP budgets differ only through the gate projection and the wider $QK^\top$ contraction.

4.3 Tasks and tiers

Main: byte-level enwik8. The first 10^8 bytes of English Wikipedia, split 90/5/5 into train/validation/test by byte offset, with a 256-symbol byte vocabulary and no tokenizer. Model: $d = 256$, 6 layers, 8 heads, $d_h = 32$, context 512, batch 32, 8000 steps, seeds $\{1, 2\}$, all six variants (48 runs). We report validation bits per character (bpc), computed as the mean cross-entropy over 50 fixed validation batches divided by $\ln 2$.

Sanity tier: character-level Shakespeare. The tiny-Shakespeare corpus at character level, 90/10 train/validation split, $d = 128$, 4 layers, 4 heads, context 256, batch 24, 3000 steps, seed 1. This tier exists to catch pipeline regressions and to give a second, smaller corpus; it is not the headline result.

Synthetic probes. Three tasks isolate positional competence from language modelling, all with a 65-symbol vocabulary (64 content symbols plus a separator) and a $d = 64$, 2-layer, 2-head model ($d_h = 32$), batch 32, 2500 steps, seeds $\{1, 2, 3\}$.

- *Offset recall*, $k \in \{8, 32\}$, context 128: a uniformly random sequence; at every position $t \geq k$ the target is the symbol at $t - k$. Solving it requires reading off an exact relative offset, and $k = 8$ versus $k = 32$ separates short from medium range.
- *Copy*, $L = 64$, context 129: the sequence is a random prefix of length L , a separator, then the same prefix repeated; only the repeated half is scored.

Accuracy is measured at scored positions on freshly sampled evaluation batches, so there is no memorizable evaluation set.

Length extrapolation. Every run is evaluated after training at $1\times$, $2\times$, and $4\times$ its training context with no adaptation, retuning, or frequency rescaling. For the LM tiers this means validation bpc at contexts 512, 1024, and 2048; for offset recall the offset k is held fixed and the sequence is lengthened; for copy the prefix length L scales with the context. Evaluation batch size is divided by the multiplier so that tokens per evaluation batch stay roughly constant.

4.4 Optimization

AdamW with $\beta = (0.9, 0.95)$, weight decay 0.1 applied to matrix-valued parameters only (no decay on norm gains, biases, or λ_h), gradient-norm clipping at 1.0, 100 steps of linear warmup, then cosine decay to a floor of 3×10^{-4} . Peak learning rate is 10^{-3} on enwik8 and 3×10^{-3} on the smaller Shakespeare and synthetic tiers. Dropout is 0. Blocks are pre-norm with RMSNorm, an MLP of width $4d$ with GELU, fused QKV projection, and no learned absolute position embeddings; output projections are scaled by $1/\sqrt{2L}$ at initialization and the output head is tied to the token embedding. Seeding is explicit for both PyTorch and NumPy, and a repeated run with the same seed reproduces the loss exactly.

4.5 Compute and reproducibility

All training runs in Modal containers: A10G GPUs for the enwik8 tier, T4 for the sanity and synthetic tiers. Each run writes a JSON record containing its full configuration, parameter count, evaluation curve, final metrics, extrapolation sweep, and wall time; these records are committed to the repository, so every number in Section 5 is traceable to a specific run. Unit tests verify the properties the method claims: that the doubled-dimension form matches the reference logit, that SWAG with gates pinned to 1 and $\lambda = 1$ reproduces ROPE logits and with gates pinned to 0 reproduces NOPE logits, and that shift consistency holds for ROPE, LEARNED-FREQ, and SWAG while failing for CAROPE.

Scale. These are approximately 5M-parameter models trained for 8000 steps. That is far below the scale at which positional mechanisms are usually deployed, and no claim here should be read as established at production scale. The purpose of this tier is a fully compute-matched comparison in which every variant is trained under an identical budget with multiple seeds, rather than a single large run whose differences cannot be attributed. We return to this limitation in Section 6.

5 Results

On enwik8, SWAG attains the best validation bits per character of all six variants at the training context on both seeds (1.479 mean against 1.499 for RoPE, a gap roughly the size of RoPE’s own seed spread), and it degrades less than RoPE at every longer evaluation context. It does not, however, extrapolate as well as learned per-head frequencies, which trade 0.02 bpc at the training context for much flatter degradation. The synthetic probes complete the picture: the variants that modify RoPE’s frequencies (LEARNED-FREQ, CAROPE) lose exact-offset generalization at longer sequences, while SWAG matches standard RoPE’s perfect accuracy everywhere. SWAG is thus the only variant tested that improves on RoPE in distribution without giving up any of RoPE’s relative-position robustness. No variant we tested is best at everything.

5.1 Language modelling on enwik8

Table 2 and Figure 2 give the main comparison. At the 512-token training context the ordering is SWAG (1.479) < GATED-SIMPLE (1.488) < ROPE (1.499) $\approx$ LEARNED-FREQ (1.500) < NOPE (1.627) $\ll$ CAROPE (2.794). SWAG improves on RoPE by 0.020 bpc on the seed means; SWAG’s own seed spread is 0.002 and RoPE’s is 0.017, so the direction of the gap is consistent, though its size is comparable to RoPE’s seed-to-seed variation and should be read accordingly at this scale.

Under length extrapolation every fixed-frequency variant degrades substantially, standard RoPE worst among the non-CAROPE variants (1.50 $\rightarrow$ 2.34 $\rightarrow$ 3.58 bpc at $1\times/2\times/4\times$). SWAG softens but does not solve this (1.48 $\rightarrow$ 2.10 $\rightarrow$ 3.25). Learned per-head frequencies extrapolate far better (1.50 $\rightarrow$ 1.75 $\rightarrow$ 2.31): freed from the fixed bank, the model can shift its channels toward longer wavelengths that remain informative past the training length. Notably, NOPE also extrapolates better than RoPE at $2\times$ (2.04 against 2.34), consistent with prior observations that decoder-only models without positional encodings degrade gracefully.

The two gated variants separate cleanly under extrapolation even though they tie in distribution: GATED-SIMPLE (1.49 $\rightarrow$ 2.48 $\rightarrow$ 3.68) tracks RoPE’s degradation, while SWAG holds a 0.3–0.4 bpc advantage over it at longer contexts. Gating the positional correction specifically, rather than scaling whole rotated channels, is where the extrapolation benefit comes from.

Our CAROPE re-implementation failed to train competitively (2.79 bpc, against 1.63 for a model with no positional information at all), and its broken shift consistency shows in the steepest extrapolation curve ($\rightarrow$ 4.63 at $4\times$). We report this as a property of our approximate re-implementation at this scale, not of Veisi et al. (2025), whose published configuration differs in details we did not replicate.

On the much smaller Shakespeare corpus (Table 3, 1.1M characters against enwik8’s 100M bytes, single seed) SWAG’s in-distribution advantage disappears: it trails RoPE by 0.03 bpc at the training context while still extrapolating better than RoPE. A plausible reading is that the gate network, which must learn its token-conditioned structure from data, needs more data than this tier provides; we flag it as an open question rather than a conclusion, given the single seed.

Table 2: Byte-level enwik8 validation bits per character, mean over seeds $\{1, 2\}$ (seed range in parentheses), at the 512-token training context and under length extrapolation to $2\times$ and $4\times$ with no adaptation. Lower is better.

Variant	Evaluation context		
	512 ($1\times$)	1024 ($2\times$)	2048 ($4\times$)
NOPE	1.627 (.001)	2.040 (.004)	3.533 (.043)
ROPE	1.499 (.017)	2.338 (.203)	3.581 (.371)
LEARNED-FREQ	1.500 (.003)	1.752 (.081)	2.311 (.080)
CAROE	2.794 (.051)	3.687 (.406)	4.627 (.332)
GATED-SIMPLE	1.488 (.002)	2.477 (.005)	3.680 (.076)
SWAG	1.479 (.002)	2.099 (.055)	3.246 (.236)

Table 3: Character-level Shakespeare sanity tier, seed 1: validation bits per character at the 256-token training context and under extrapolation. Included as a second corpus and a pipeline check, not as a headline result.

Variant	256 ($1\times$)	512 ($2\times$)	1024 ($4\times$)
NOPE	2.313	2.592	3.567
ROPE	2.189	2.707	3.697
LEARNED-FREQ	2.182	2.440	2.821
CAROE	2.340	4.148	5.271
GATED-SIMPLE	2.196	2.868	3.944
SWAG	2.218	2.587	3.429

5.2 Synthetic positional probes

In distribution (Table 4), every variant with positional information solves both offset-recall tasks and the copy task exactly, while NOPE is at chance-like levels on offset recall and partially solves copy through content alone. The probes therefore discriminate only under length extrapolation (Table 5), where the picture is sharp: standard RoPE and SWAG preserve offset-recall accuracy essentially perfectly out to $4\times$ the training length on every seed (≥ 0.994 , SWAG at 1.000 on all six task/seed combinations), while the variants that modify the frequency bank do not. LEARNED-FREQ falls to 0.732 mean at $k=32$, $4\times$, with severe seed variance (two seeds near 0.5–0.65, one intact), and CAROE falls as low as 0.499 mean at $k=8$. The same mechanism that helps those variants extrapolate on natural text (moving energy away from the fixed bank) destroys the exact relative geometry the offset task requires. SWAG, which cannot move the bank but only reweight it, inherits RoPE’s robustness unchanged.

Copy under extrapolation fails for every variant (≈ 0.016 , chance) once the required attention offset (L , scaled with context) exceeds anything seen in training; position encodings alone do not induce length-generalizing copying in these small models, consistent with copy generalization being an algorithmic rather than positional failure.

5.3 What the gates learn

We take the seed-1 SWAG enwik8 checkpoint, run validation batches with gate recording enabled, and average the gate value $g_{m,h,i}$ over tokens grouped by the character class of the byte at position

Table 4: Synthetic probes: accuracy at scored positions at the training length, mean over seeds $\{1, 2, 3\}$ (seed range in parentheses). Offset recall predicts the symbol k positions back; copy reproduces a length-64 prefix after a separator.

Variant	offset $k=8$	offset $k=32$	copy $L=64$
NOPE	0.064 (.004)	0.045 (.003)	0.463 (.032)
ROPE	1.000 (.000)	1.000 (.000)	1.000 (.000)
LEARNED-FREQ	1.000 (.000)	1.000 (.000)	1.000 (.000)
CARPE	1.000 (.000)	1.000 (.000)	1.000 (.000)
GATED-SIMPLE	1.000 (.000)	1.000 (.000)	1.000 (.000)
SWAG	1.000 (.000)	1.000 (.000)	1.000 (.000)

Table 5: Synthetic probes under length extrapolation: accuracy at $2\times$ and $4\times$ the training sequence length, offset k held fixed, copy prefix length scaled with context. Mean over seeds $\{1, 2, 3\}$; bold marks the fixed-geometry variants that remain solved.

Variant	offset $k=8$		offset $k=32$		copy $L=64$	
	$2\times$	$4\times$	$2\times$	$4\times$	$2\times$	$4\times$
NOPE	0.059	0.047	0.033	0.026	0.025	0.014
ROPE	1.000	0.994	1.000	1.000	0.016	0.015
LEARNED-FREQ	1.000	0.998	0.843	0.732	0.016	0.015
CARPE	0.718	0.499	0.996	0.806	0.016	0.016
GATED-SIMPLE	1.000	1.000	1.000	0.998	0.016	0.015
SWAG	1.000	1.000	1.000	1.000	0.016	0.015

m (letter, digit, whitespace, punctuation, newline; 81,920 tokens), separately per layer, head, and frequency index. Figure 1 shows per-layer profiles averaged over heads; Table 6 gives the learned per-head λ_h .

The gates do not collapse to the endpoints: individual (head, channel, class) means span 0.005 to 1.000, and head-averaged values span 0.26–0.99, so the model uses the full interpolation between “position off” and “full RoPE.” Three regularities stand out. First, structure varies along both axes, class and frequency, and changes with depth. In layer 0, digits suppress the most local channel far below other classes (head-averaged $g = 0.42$ at $i=0$, against 0.82 for whitespace), while whitespace keeps nearly all channels open (class mean 0.91); a byte inside a digit string is locally near-interchangeable, and the model learns to mute exactly the channels that encode fine local order. Second, middle layers develop a shared scale preference modulated by class: in layer 3 every class peaks at the same mid-band channel ($i=9$; 0.89–0.99) while long-wavelength channels ($i \geq 12$) are turned down to 0.49–0.61. Third, mean gate values decline with depth (layer-0 class means 0.75–0.91, layer-5 means 0.57–0.73, lowest for newline and whitespace), i.e. deeper layers rely progressively less on positional correction, most strongly for delimiter-like tokens. Meanwhile every learned λ_h but one ends above its initialization of 1.0 (range 0.70–1.52, mean 1.29): the model amplifies the positional scale overall while the gates prune it selectively per token.

Cost. At the enwik8 tier SWAG averaged 908s of wall clock per run against 704s for RoPE on the same A10G configuration, a $1.29\times$ overhead. This reflects both the doubled $QK^\top$ contraction width and the fact that the mismatched query/key versus value dimensions exclude the fastest fused

Table 6: Learned per-head positional scale λ_h in the seed-1 SWAG enwik8 model. Initialized at 1.0 for every head; all but one head increase it.

Layer	h_0	h_1	h_2	h_3	h_4	h_5	h_6	h_7
0	1.21	1.23	1.26	1.27	1.20	1.36	1.26	1.21
1	1.32	1.21	1.36	1.49	1.30	1.32	1.22	1.35
2	1.44	1.22	1.37	1.18	1.39	1.40	1.50	1.39
3	1.39	1.31	1.35	1.36	1.28	1.43	1.03	1.52
4	1.04	1.39	1.35	0.70	1.24	1.04	1.43	1.21
5	1.14	1.47	1.24	1.38	1.47	1.28	1.36	1.18

attention kernel in this PyTorch version; GATED-SIMPLE, which keeps matched dimensions, ran at $1.13\times$ RoPE. A fused kernel for the doubled form would close much of the gap; we did not write one.

6 Discussion

What the comparison establishes. The results form a three-way trade rather than a single winner. Standard RoPE offers exact relative geometry and perfect synthetic-offset generalization but the worst natural-text extrapolation of the healthy variants. Learned per-head frequencies buy the best natural-text extrapolation by moving the frequency bank, and pay for it exactly where the theory predicts: exact-offset recall collapses on two of three seeds once sequences exceed the training length. SWAG occupies the remaining corner: best in-distribution bits per character, extrapolation consistently better than RoPE though worse than LEARNED-FREQ, and RoPE’s synthetic robustness preserved to the last decimal place. Which corner matters depends on the deployment; what the study rules out, at this scale, is the assumption that input-dependent positional flexibility is free.

Attributing the in-distribution gain. The gated variants carry 4.1% more parameters than ROPE (Table 1), and GATED-SIMPLE, parameter-matched to SWAG to within 50 parameters, also improves on ROPE at the training context (1.488 against 1.499). Some of the in-distribution gain is therefore plausibly gate capacity rather than the correction-gating mechanism specifically, and we do not claim otherwise. What parameter count cannot explain is the extrapolation separation between the two gated variants: SWAG holds a 0.3–0.4 bpc advantage over GATED-SIMPLE at $2\times$ and $4\times$ context with the same parameters, the same gate inputs, and the same geometry, differing only in whether the gate multiplies the positional correction or the whole rotated channel. That contrast is the cleanest evidence in the study that gating the correction is the operative choice.

Why gating helps at longer contexts. A fixed learned gate profile that suppresses long-wavelength channels (Section 5.3, layer 3) resembles a learned, token-conditioned version of post-hoc frequency rescaling (Peng et al., 2024), which suppresses or rescales the channels that alias beyond the training length using a hand-designed, token-independent schedule. We did not test whether SWAG’s gate values shift when the same token is evaluated at longer contexts, so we cannot distinguish “the gates adapt” from “a fixed profile happens to prune the fragile channels”; the second explanation is simpler and consistent with everything we measured.

The relativity trade, tested directly. CAROPE’s extrapolation columns are the empirical form of the argument in Section 3.2: token-conditioned frequencies make the logit depend on absolute positions, and at $4\times$ context its offset-recall accuracy falls to 0.50–0.81 while SWAG, with the same parameter budget and the same conditioning signal, stays at 1.000. This holds even though our CAROPE re-implementation trained poorly on natural text, because the probes are solved to accuracy 1.000 by every positional variant in distribution; the failure appears only where relativity is the binding constraint.

Reading the gates. The gate structure is systematic along character class, frequency, and depth, and it is a correlational reading of a single checkpoint. We note it because the structure is the kind the mechanism was designed to allow (digits muting local order, delimiters shedding positional correction at depth), not because one checkpoint can establish that the structure causes the quality difference. Intervention experiments, such as clamping the gates of one class to their initialization, would be the next step.

6.1 Limitations

The scale is small. All models are approximately 5M parameters trained for 8000 steps on 512-token contexts, three orders of magnitude below where positional encodings are usually contested. Effects that appear here may not survive scaling, and effects that matter at scale may be invisible here. The compensating design choice is that every variant is trained under an identical budget with multiple seeds, so within-study comparisons are controlled even though the absolute numbers are not competitive.

Seed coverage is thin. The main tier uses two seeds per variant and the synthetic probes three. We report the seed spread alongside every mean and do not run significance tests, which two seeds cannot support. Differences smaller than the observed spread should be read as ties.

CAROPE is our re-implementation, following the description of token-conditioned head-specific frequencies in Veisi et al. (2025) rather than the authors’ code, and adapted to this architecture and scale. It should be read as a representative of the input-dependent-frequency family under matched conditions, not as a reproduction of the published results, and a poor showing here is evidence about the mechanism in this setting rather than about that paper.

Both language-modelling tiers are character- or byte-level. Byte-level modelling changes the relationship between token index and semantic distance: a 512-byte context is a short passage, and the frequency channels that matter for bytes are not obviously the ones that matter for subword tokens. Whether the gate structure reported in Section 5.3 has an analogue under BPE is untested. Similarly, we evaluate extrapolation only to $4\times$ the training context, far short of the regimes that motivate methods such as YaRN.

Finally, the comparison is architecturally narrow: one pre-norm decoder-only design, one optimizer configuration, one learning-rate schedule per tier. We did not tune hyperparameters per variant, which is the correct choice for a matched comparison but means each variant is evaluated at settings chosen for the family rather than for itself.

6.2 Reproducibility and ethics

All code, configurations, and per-run JSON records are public under the MIT license at <https://github.com/jamesbaker1/swag-rope>. Every number in Section 5 is traceable to a committed run record containing the exact configuration, parameter count, evaluation curve, extrapolation sweep, and wall time that produced it. All training and evaluation ran in Modal containers (A10G

for the main tier, T4 for the sanity and synthetic tiers) from the same container image, so the environment is pinned rather than described. The total compute for the full experiment matrix is approximately 3.0 GPU-hours across all 78 runs (A10G and T4 combined), summed from the `wall_time` fields of the run records.

The work uses two public datasets, enwik8 (English Wikipedia text) and tiny-Shakespeare, plus synthetic sequences; no human subjects, personal data, or annotator labor are involved. The contribution is a component-level change to attention with no application attached, and we see no misuse pathway specific to it beyond those already present in language modelling generally.

References

- Riccardo Ali, Alessio Borgi, Christopher Irwin, Mario Severino, and Pietro Liò. Remember to forget: Gated adaptive positional encoding. *arXiv preprint arXiv:2605.10414*, 2026. URL <https://arxiv.org/abs/2605.10414>.
- Olga Golovneva, Tianlu Wang, Jason Weston, and Sainbayar Sukhbaatar. Contextual position encoding: Learning to count what’s important. *arXiv preprint arXiv:2405.18719*, 2024. URL <https://arxiv.org/abs/2405.18719>.
- Amirhossein Kazemnejad, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Payel Das, and Siva Reddy. The impact of positional encoding on length generalization in transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL <https://arxiv.org/abs/2305.19466>.
- Sajad Movahedi, Timur Carstensen, Arshia Afzal, Frank Hutter, Antonio Orvieto, and Volkan Cevher. Selective rotary position embedding. *arXiv preprint arXiv:2511.17388*, 2025. URL <https://arxiv.org/abs/2511.17388>.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2309.00071>.
- Ofir Press, Noah A. Smith, and Mike Lewis. Train short, test long: Attention with linear biases enables input length extrapolation. In *International Conference on Learning Representations (ICLR)*, 2022. URL <https://arxiv.org/abs/2108.12409>.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL <https://arxiv.org/abs/1910.10683>.
- Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*, 2021. URL <https://arxiv.org/abs/2104.09864>.
- Ali Veisi, Delaram Fartoot, and Hamidreza Amirzadeh. Context-aware rotary position embedding. *arXiv preprint arXiv:2507.23083*, 2025. URL <https://arxiv.org/abs/2507.23083>.

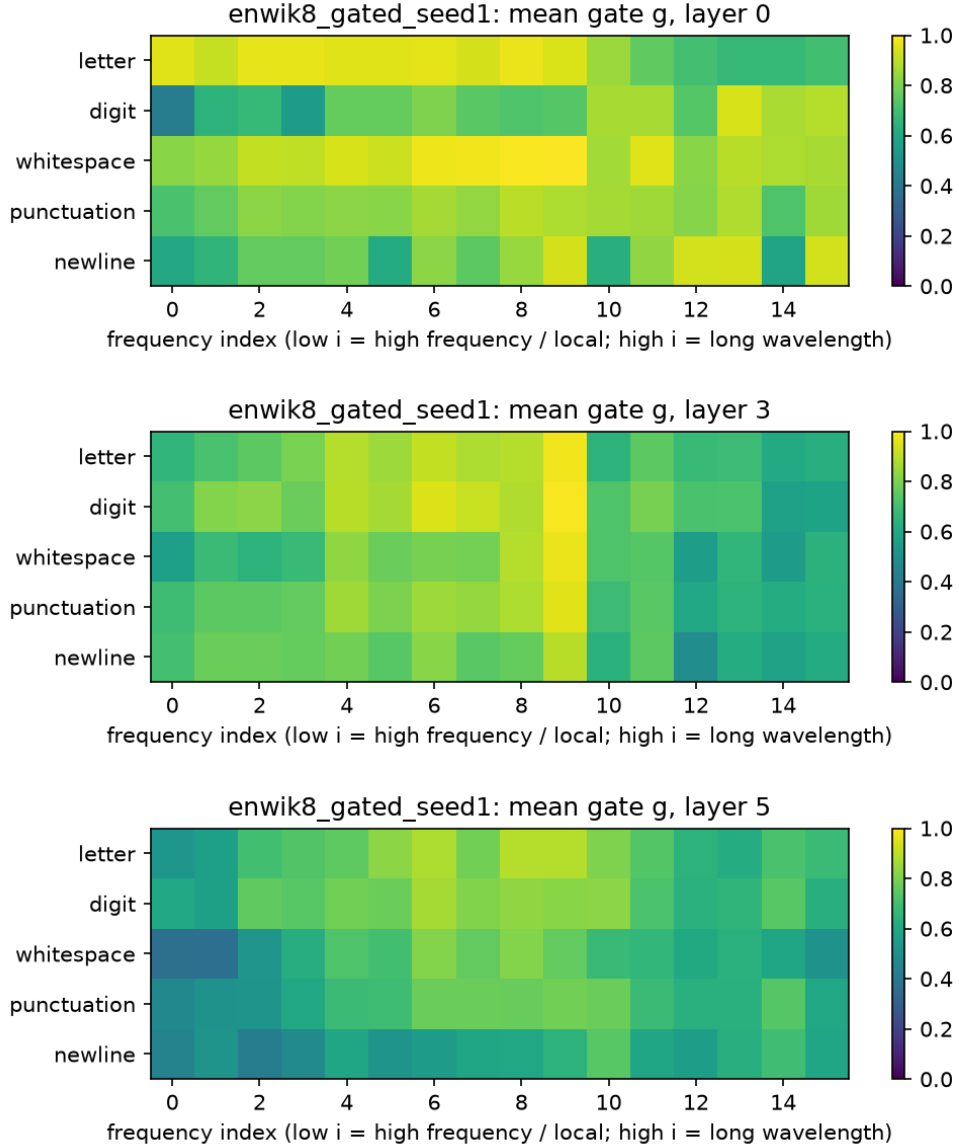


Figure 1: Mean SWAG gate value by character class and frequency channel (averaged over heads) for layers 0, 3, and 5 of the seed-1 enwik8 model. Low frequency index corresponds to short-wavelength (local) channels, high index to long-wavelength channels.

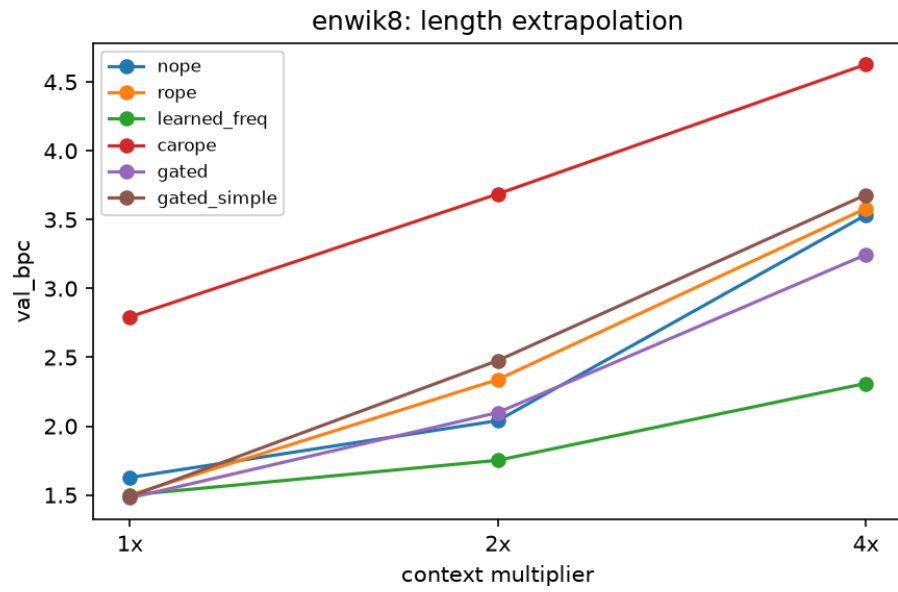


Figure 2: Validation bits per character against evaluation context length on enwik8, all variants, no adaptation at longer contexts. Mean over seeds.